MODULE *DieHard*

EXTENDS *Integers*

VARIABLES x, y

$$Init \triangleq \land (x = 0) \\ \quad \land (y = 0)$$

$$FillSmall \triangleq \land (x' = 3) \\ \quad \land (y' = y)$$

$$FillBig \triangleq \land (x' = x) \\ \quad \land (y' = 5)$$

$$EmptySmall \triangleq \land (x' = 0) \\ \quad \land (y' = y)$$

$$EmptyBig \triangleq \land (x' = x) \\ \quad \land (y' = 0)$$

$$\begin{aligned} PourFromSmall \triangleq\ & \lor (\land (x \leq 5 - y) \\ & \quad \land (x' = 0) \\ & \quad \land (y' = x + y)) \\ & \lor (\land (x > 5 - y) \\ & \quad \land (x' = x - 5 + y) \\ & \quad \land (y' = 5)) \end{aligned}$$

$$\begin{aligned} PourFromBig \triangleq\ & \lor (\land (y \leq 3 - x) \\ & \quad \land (x' = x + y) \\ & \quad \land (y' = 0)) \\ & \lor (\land (y > 3 - x) \\ & \quad \land (x' = 3) \\ & \quad \land (y' = y - 3 + x)) \end{aligned}$$

$$\begin{aligned} Next \triangleq\ & \lor FillSmall \\ & \lor FillBig \\ & \lor EmptySmall \\ & \lor EmptyBig \\ & \lor PourFromSmall \\ & \lor PourFromBig \end{aligned}$$

$$TypeOK \triangleq \land (x \in 0 \,..\, 3) \\ \quad \land (y \in 0 \,..\, 5)$$

* Modification History

* Last modified Sun *Apr* 12 10:44:26 *CST* 2026 by *digogonz*

\ * Created Sun *Apr* 12 09:24:29 *CST* 2026 by *digogonz*